SmartHub Diagnostics â DFD (Text Only) â Level 0 (Context)

Method: Gane and Sarson DFD notation

State (In this level): SmartHub is one process (P0) with external entities and top-level data flows.
Define (According to): Context DFD shows the entire system as one process and only external interfaces (Gane & Sarson, 1979).
Justify (Moreover): This gives a clear and bounded overview before decomposition.

Process:
P0 SmartHub Diagnostics (UI + Node/Express Backend)

External Entities:
- E1 Technician
- E2 Android Phone
- E3 Windows Host OS (USB/PnP/MTP evidence)
- E4 ADB / Fastboot Tools
- E5 Camera Checker (optional visual check)
- E6 Offline AI Helper (Python)

Data Store:
- D1 Storage (JSON + SQLite)

Main Data Flows:
- E1 -> P0: request / start
- P0 -> E1: results / report
- E2 -> P0: device signals
- E3 -> P0: USB/PnP evidence
- E4 -> P0: adb/fastboot output
- P0 -> E5: capture request (optional)
- P0 -> E6: suggest request; E6 -> P0: AI conclusion (optional)
- P0 -> D1: write results; D1 -> P0: read history/config

How to read (flow)

Level 0 shows SmartHub as one process and lists external interfaces in Gane and Sarson style.
This page is text-only; the diagram PDF shows the same elements graphically.

SmartHub Diagnostics â DFD (Text Only) â Level 1 (Major Processes)

Method: Gane and Sarson DFD notation

State (In this level): P0 is decomposed into major processes and linked stores.
Define (According to): Level 1 decomposition must keep parent-level flow balance (Gane & Sarson, 1979).
Justify (Moreover): SmartHub needs this level to separate and clarify major diagnostic functions.

Processes (decomposition of P0):
- P1 Connection Check
- P2 Full Diagnostics (ADB)
- P3 BSOD Diagnose (USB-only)
- P4 Camera Visual Check
- P5 Security Scan (ADB)
- P6 Offline AI Suggest

Data Stores:
- D1 history.json (run summaries / history)
- D4 memory.sqlite (cases) â offline AI memory
- D5 adb_ai_memory.sqlite (adb_cases) â ADB AI memory

Key Data Flows:
- P1 -> D1: save check result
- P2 -> D1: save run
- P3 -> D1: save BSOD report
- P3 -> D4: write offline AI case (optional helper)
- P5 -> D1: save security report
- P6 <-> D4: read similar cases / write offline AI case
- P6 <-> D5: read ADB memory / write ADB AI case

How to read (flow)

Level 1 breaks P0 into major processes and shows which stores they read/write.
Only primary stores are listed here; the graphical DFD shows the same relationships as arrows.

SmartHub Diagnostics â DFD (Text Only) â Level 2 (P3 BSOD Diagnose â USB-only)

Method: Gane and Sarson DFD notation

State (In this level): P3 is decomposed into sub-processes P3.1-P3.5.
Define (According to): Level 2 expands one selected Level 1 process for detailed transformation flow (Gane & Sarson, 1979).
Justify (Moreover): This level is needed to make the USB-only BSOD decision path auditable and clear.

Decomposed Process:
P3 BSOD Diagnose (USB-only) decomposes into:
- P3.1 Detect USB-connected device
- P3.2 Collect USB/BSOD evidence (logs)
- P3.3 Camera visual check (optional)
- P3.4 Generate offline AI suggestion (optional)
- P3.5 Build + save diagnostic result

External Entities used by P3:
- E1 Technician
- E3 Windows Host OS (USB/PnP/MTP evidence)
- E5 Camera Checker (optional)
- E6 Offline AI Helper (Python, optional)

Data Stores used by P3:
- D1 history.json (save run summary)
- D4 memory.sqlite (cases) â save offline AI memory
- D7 usb-evidence files (save host logs/evidence)

Key Data Flows:
- E1 -> P3.1: start USB-only BSOD
- E3 -> P3.1/P3.2: USB enumerate + logs
- E5 -> P3.3: camera frames (optional)
- P3.2 -> D7: save USB evidence
- P3.4 -> D4: save AI memory (optional)
- P3.5 -> D1: save run summary
- P3.5 -> E1: result summary

How to read (flow)

Level 2 decomposes one Level 1 process (P3) into its sub-steps and shows the main stores used by those steps.
USB-only means this path can work even when ADB debugging is not available.